

BIOL 220 Problem Set 03: Sample estimates and confidence intervals

Answer Key

How much caffeine is in your coffee?

The table below (not shown in answer key) contains data on the amount of caffeine in a 16 oz. cup of coffee obtained from various vendors. For context, doses of caffeine over 25 mg are enough to increase anxiety in some people, and doses over 300 to 360 mg are enough to significantly increase heart rate in most people. A can of Red Bull contains 80mg of caffeine.

Use the data table of coffee caffeine concentration to answers Questions 1–6

1. What is the mean amount of caffeine in the sample of 16 oz. coffees? Round your answer to 2 decimal places. [1 point]

i Answer

```
ybar <- mean(caf$caffeine_mg_16oz)
ybar
```

```
[1] 188.0643
```

2. What is the 95% confidence interval for the mean? Use the 2 SE rule and round your answer to 2 decimal places. Report your answer like “lower - upper”. For example, if the confidence interval were -1.11 to 2.22, then I would report “-1.11, - 2.22”. [2 point]

i Answer

```
se <- sd(caf$caffeine_mg_16oz) / sqrt(nrow(caf))  
  
ci95 <- c(ybar - 2 * se, ybar + 2 * se)  
ci95
```

```
[1] 169.0485 207.0801
```

3. Is the amount of caffeine in a cup of coffee relatively consistent from one vendor to another? To help answer that, calculate the standard deviation of caffeine level and round your answer to 2 decimal places. [1 point]

i Answer

```
sd(caf$caffeine_mg_16oz)
```

```
[1] 35.57535
```

With a standard deviation of 35.58, the amount of caffeine is not relatively consistent

4. The table below (not shown in answer key) has data on six 16 oz. cups of Breakfast Blend coffee sampled on six different days from a Starbucks location. Calculate the the 95% confidence interval for the mean using the 2 SE rule for these data. Report your answer in the same format as Question 2. [2 point]

i Answer

```
# file path will be different for you  
bucks <- read.csv("../data/caffeineStarbucks.csv")  
  
se <- sd(bucks$caffeine_mg_16oz) / sqrt(nrow(bucks))  
  
ci95 <- mean(bucks$caffeine_mg_16oz) + 2 * c(-se, se)  
ci95
```

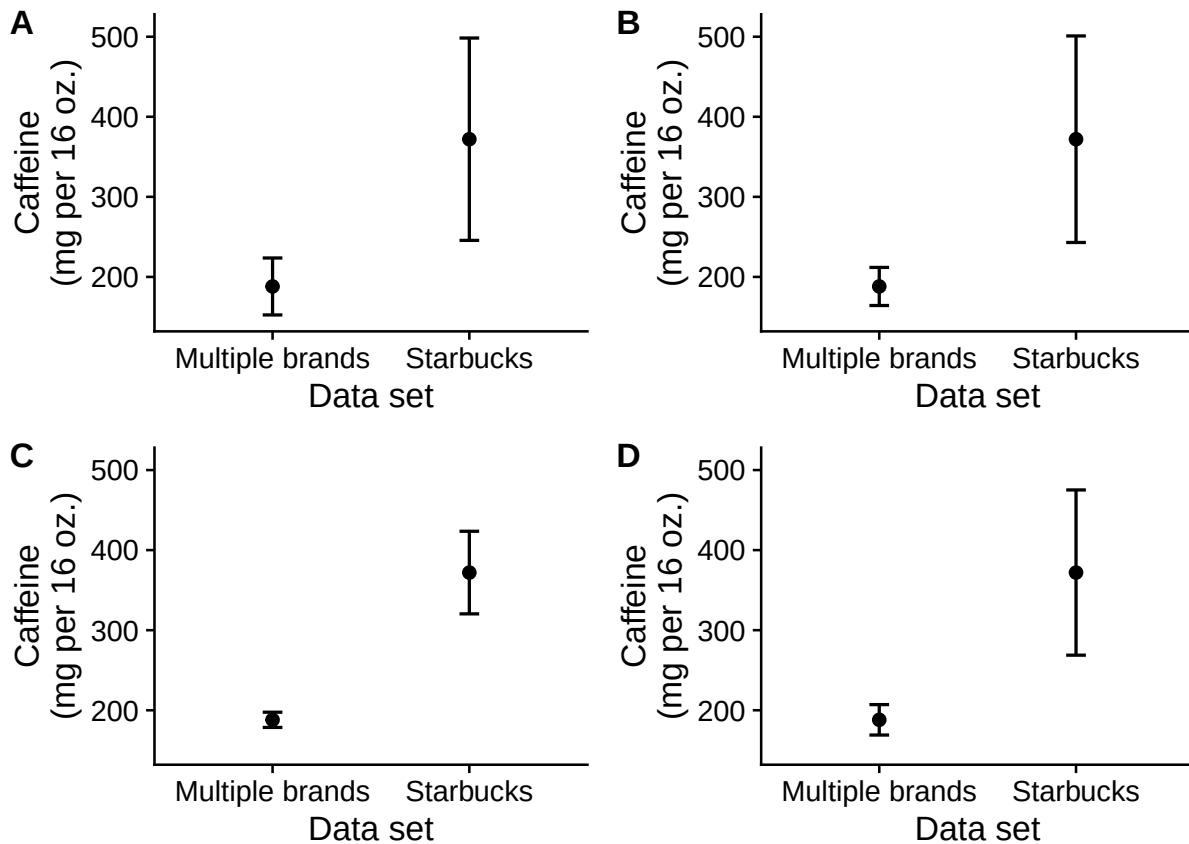
```
[1] 268.7885 475.1448
```

5. Compare these results to the data taken on the broader sample of vendors in the first file. Describe the difference in 1-2 sentences. [2 point]

i Answer

The range of the 95% CI for Starbucks is substantially bigger (spanning roughly 200 mg per 16 oz) compared to the 95% CI from multiple brands (which spans roughly 40 mg per 16 oz). Two factors could contribute to this difference in the widths of the CIs: there are fewer data points for Starbucks, there could also be a genuinely huge discrepancy in how much caffeine is in the Starbucks coffee.

6. Based on the two data sets above (Multiple brands and Starbucks), which of the figures below depicts the mean caffeine concentration with error bars equal to ± 1 standard error of the mean? [1 point]



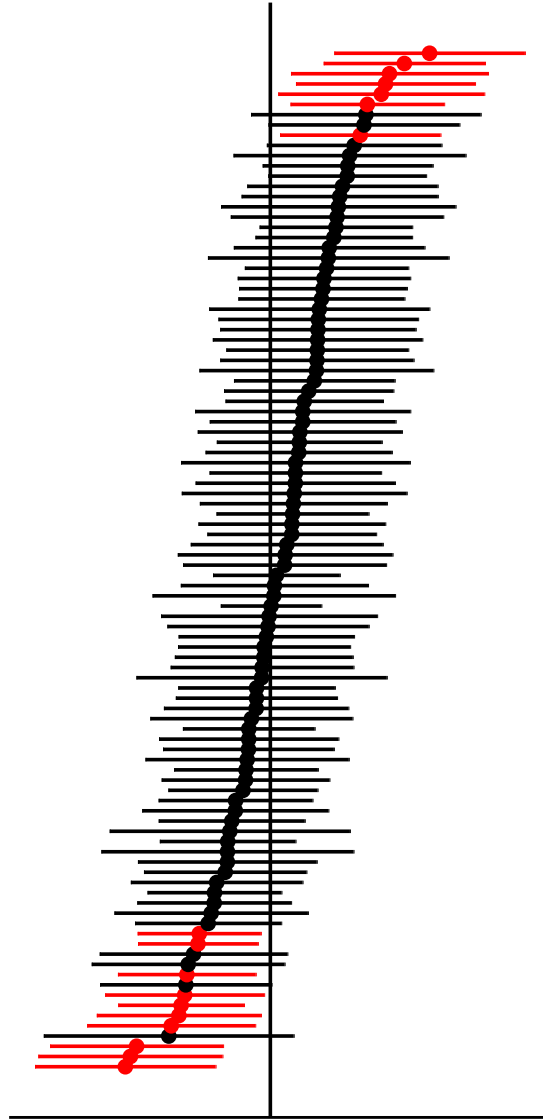
i Answer

C.

Understanding confidence intervals

The figure below shows sample means (points) and associated confidence intervals of the mean (horizontal lines) from 100 different samples of the same population. In each case, the sample size is exactly $n = 16$. Assume that all samples were conducted using an identical procedure.

83 of 100 have confidence intervals that overlap the true population mean (vertical black line). These are shown with **black** points and lines. 17 of 100 have confidence intervals that do **not** overlap the true population mean (vertical black line). These are shown with red points and lines.



💡 Use the figure above to answer Question 9

7. Which of the following statements most likely explains what you observe in this figure? [1 point]
- These are 95% confidence intervals from 100 random samples
 - These are 95% confidence intervals from 100 nonrandom samples
 - These are 95% confidence intervals but some confidence intervals do not overlap the population mean because the sample size is too small
 - These are 95% confidence intervals but but some confidence intervals do not overlap

the population mean because the samples are biased

Answer

The samples themselves genuinely are random – `rep_slice_sample()` draws each sample uniformly at random from the population – so “nonrandom samples” is not correct.

The real explanation is in the code that builds this figure: after drawing each random sample of size 16, it duplicates every value onto itself (`bind_rows(. , .) / rbind(x, x)`, commented “this step simulates pseudo-replication”) before calculating the standard error. That step doubles the apparent sample size from 16 to 32 without adding any real, independent information, which under-estimates the standard error and makes each interval narrower than a genuine 95% CI should be. Simulating this exact procedure shows this pushes the miss rate from the expected ~5% up to somewhere around 15-20% of intervals failing to cover the true mean.

Note to instructor: none of the four answer choices describes this mechanism precisely. “The sample size is too small” is the closest of the four in spirit (the standard error is calculated as if there were more independent data than there really are), but you may want to revise the options to name pseudo-replication/non-independence directly as the cause.